

Software Requirements Specification (SRS)

6-Platform eBay Helpdesk

1. Project Overview and Goals

Mission Statement

This document defines the requirements for a centralized customer service communication hub designed to aggregate and manage eBay-specific interactions across six internal business units: **Main, Marine, Tech, Ship, Tools, and Trade**. The system is engineered to provide a unified interface for high-volume message handling while ensuring strict adherence to response protocols.

Scope Definition

The system architecture is strictly limited to eBay Marketplace API integrations. The ingestion engine is dedicated to fetching and synchronizing messages from eBay accounts only; third-party email providers (e.g., Gmail, Outlook, Yahoo) and other external CRM integrations are explicitly out of scope for this version.

Business Objectives

- **Centralized Visibility:** Consolidate disparate eBay account messages into a single, high-performance dashboard for cross-platform brand management.
- **AI-Driven Operational Efficiency:** Automate the categorization of incoming traffic to reduce manual triage time and prioritize urgent customer needs.
- **SLA Compliance & Risk Mitigation:** Maintain a strict 24-hour response window for all eBay communications to preserve seller ratings and marketplace standing.

2. User Roles and Permissions

The system implements a granular Role-Based Access Control (RBAC) model to ensure data integrity and operational security.

Role	Primary Responsibilities	Permissions & Technical Access
Admin	System configuration, account orchestration, and user lifecycle management.	Manage eBay API credentials/webhooks; Configure AI confidence thresholds; Manage Operation Manager accounts; Access system-wide audit logs.
Operation Manager	Front-line message handling, ticket lifecycle management, and resolution.	View, assign, and reply to tickets; Apply manual tags; Manage personal "My Tickets" queue; Utilize @mentions for internal collaboration.

3. Functional Requirements: Unified Inbox & Communication

Email Ingestion & Background Processing

The backend must implement a robust, asynchronous fetching mechanism to poll the eBay API for the six specified platforms.

- **Asynchronous Task Queue:** To prevent blocking the FastAPI event loop, the system shall utilize **Celery with Redis** as a message broker to manage high-frequency polling and webhook processing across all integrated eBay channels.

Visual Interface Components (The "To Do" View)

Based on [SOURCE_IMAGE_1], the message list must include the following dynamic UI components:

1. **Platform & Branding:** Clear display of brand-specific icons (Main, Marine, etc.) and marketplace logos (e.g., Walmart/eBay logos as per source) to provide immediate channel context.
2. **Geographic Indicators:** Display of sender country flags to assist in language identification and regional shipping considerations.
3. **Customer Metadata:** Display of customer names and a "VIP" status badge for high-value account identification.
4. **"Latest Update" Column:** A dedicated column displaying the message Subject Line and a text snippet of the most recent interaction for rapid preview without opening the ticket.

AI Classification Engine

The system shall utilize a machine-learning-based classification engine to tag incoming messages automatically.

- **Required Categories:** 'Cancellation', 'Return request', 'Pre-sales', and 'Irrelevant' (per [SOURCE_IMAGE_2]).
- **Technical Thresholds:** The engine must provide a confidence score for each tag. Messages falling below a defined threshold should be flagged for manual review by an Admin.

SLA and Urgency Monitoring

To enforce the 24-hour SLA, the UI must provide dual visual indicators of urgency:

- **Time Remaining:** A "Time left" column displaying hours remaining (e.g., "6 hours left").

- **Deadline Tracking:** A "Respond Till" column featuring a color-coded progress bar (Green > 12h, Yellow 6-12h, Red < 6h) as visualized in [SOURCE_IMAGE_2].

4. Ticket Lifecycle and Workflow Management

Ticket States and Navigation

The system must support the full lifecycle of a customer inquiry. The following states, derived from [SOURCE_IMAGE_2], are mandatory:

- **Unassigned:** New messages awaiting ownership.
- **My Tickets:** Messages assigned to the authenticated user.
- **Not Resolved:** Active tickets requiring further action.
- **Resolved:** Closed tickets where the inquiry is complete.
- **Snoozed:** Tickets hidden from the main view until a specified trigger time.
- **Delayed:** Tickets flagged for extended investigation.
- **Blocked:** Tickets where further progress is halted by external dependencies.
- **Archive:** Long-term storage for historical records.

Assignment & Collaboration

- **Ownership Logic:** Every ticket must have an "Owner" or "Assigned User" field. Operation Managers can manually "Take Ownership" of unassigned tickets.
- **Internal Notes & @Mentions:** The system must support an internal-only comment thread within each ticket. Operation Managers must be able to @mention other users to trigger notifications for collaborative resolution.

5. Data Enrichment and Real-time Sidebar

Contextual Data Integration

The workspace must feature a real-time sidebar that fetches live data from the eBay API upon ticket selection to eliminate tab-switching.

Specific Data Points

- **eBay Order Details:** Item ID, quantity, and purchase date.
- **Tracking Numbers/Status:** Real-time carrier updates and delivery verification.
- **Sales Order (S/O) Values:** Display of the total transaction value. The UI must support multi-currency symbols (e.g., \$, £, €) as identified in the source metadata.

6. Technical Stack and Infrastructure

Frontend

- **Framework:** React (TypeScript).

- **Requirements:** A responsive, single-page application (SPA) architecture utilizing optimized state management to handle real-time ticker updates for the countdown timers.

Backend

- **Framework:** Python (FastAPI).
- **Logic:** High-concurrency handling of eBay API requests and AI classification logic.
- **Task Management:** Celery with a Redis backend for background job processing (polling, email fetching, and notification dispatch).

Database

- **Storage:** PostgreSQL for persistent storage of ticket data, customer metadata, and audit trails.
- **Version Control:** Alembic is mandated for all database schema migrations to ensure environment parity.

7. Filtering and Navigation

Advanced Filtering and Sorting

The dashboard header must provide comprehensive tools for managing large volumes of data:

- **Filter Categories:** Marketplaces, Brands, Channels, Message Type, AI Classification, Manual Tags, Owner, and Mentioned.
- **Message Source Logic:** Include a "Last message sent by" filter to distinguish between system-automated responses and manual user replies.
- **Sorting:** Users must be able to sort the ticket list by **Channel, Date Received, and Urgency (SLA)**.
- **Global Search:** A persistent search bar to locate tickets by Customer Name, Order ID, or Subject.

Sidebar Navigation Tree

The left-hand navigation must provide immediate access to:

- **Core Views:** Dashboard, All Tickets, Starred, and My Tickets.
- **Status Folders:** Direct navigation to Not Resolved, Unassigned, Resolved, Blocked, Snoozed, Delayed, and Archive.
- **Channel Folders:** Direct navigation to the six specific eBay platforms.